Module 7 Assignment Solution

Ans to ques no 1:

- Django is a python based web-framework that is used to create full-stack web applications. It is mainly a backend framework but it also provides features for handling the frontend, such as templates and static files.
- Here is 5 important features of Django :-
    1. Built-in admin panel
    2. MVT Achitecture
    3. URL Routing
    4. ORM (Object-Relational Mapping)
    5. Built-in Authentication System
- Difference between MVC and MVT architecture:

MVC	MVT
MVC means Model-View-Controller.	MVT means Model-View-Template
In MVC, view handles the presentation/UI.	But in MVT, template handles the UI.
Controller handles the request and application logic.	View handles the request and application logic.
Template is generally part of the View layer.	Template is a separate component.
Controller connects the Model and View.	View connects the Model and Template.

- **Model:** Model defines what data my application stores in the database. It is responsible for data and database operations.

  class Project(models.Model):

  title = models.CharField(max_length=200)

  description = models.TextField()

  We can consider the Project model as a representation of a database table. title and description are fields (columns) of that table, while each project stored in the database represents a row.

View: View handles the request and decides what response should be return. It receives the request, get data from the model, sends that data to the template and returns the response.

Template: Template is responsible for displaying the data to the user. It receives the data from the View and and displays it as HTMLl. It's mainly the User Interface of a web application.

URL: URL determines which view should handle a particular URL.
Path("home/", views.home)
If user visits home/ then Django matches the URL and calls **views.home**

Ans to ques no 2:

Here is the step-by-step process of creating a Django project:

Step 1: Create a virtual environment.

A virtual environment is an isolated Python environment where we can install project-specific packages without affecting other Python projects.

First, create a project directory and navigate into it:

mkdir MyProject

cd MyProject

Create a virtual environment:

python -m venv venv

Here, venv is the name of the virtual environment.

Activate the virtual environment:

venv\Scripts\activate

Step 2: Install Django.

pip install django

To check whether Django has been installed successfully:

django-admin –version

Step 3 : Creating a Django Project

django-admin startproject mysite .

Here:

- startproject is the command used to create a Django project.
- mysite is the project name.
- . means creating the project in the current directory instead of creating another nested directory.

Step 4 : Explaining the purpose of manage.py, settings.py, urls.py and wsgi.py:

Manage.py: It is a command-line utility that helps us interact with our Django project. We use it to run commands such as starting the development server, creating applications, applying migrations etc.

Settings.py: settings.py contains the configuration settings of a Django project. It includes database configuration, installed apps, middleware, templates, static files, security settings and other project configurations.

Urls.py: urls.py defines the URL patterns of a Django project. It maps incoming URLs to their corresponding Views so that Django knows which view should handle a particular request.

Wsgi.py: wsgi.py is the entry point for WSGI-compatible web servers to communicate with a Django application. It is mainly used when deploying a Django project in a sunchronous WSGI server environment.

Step 5 : Running the Django Development Server

After creating the project, run the following command in the terminal:

python manage.py runserver

If everything is configured correctly, Django will start its development server at :

http://127.0.0.1:8000/